

Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

$f(x, y, a, b)$

```
{  
  if (x is 1) y = a;  
  else y = b;  
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?

Select one:

- ☐ a. Multiplexer
- ☐ b. Decoder
- ☐ c. Flip Flop
- ☐ d. Full adder

How many (4 bit full adders)chipsets used to calculate the sum of $(17+18)_{10}$

Select one:

☒ a. 5

☐ b. 2

☐ c. 1

☐ d. 3

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How many NAND gates (optimally) used to design the function :

$$F = \bar{A}B + A\bar{C}$$

Select one:

- ☐ a. 3
- ☐ b. 7
- ☐ c. None of them
- ☐ d. 4

What is the largest number of data inputs which a data selector with two control inputs can have ?

Select one:

☐ a. 8

☐ b. 16

☐ c. 2

☐ d. 4

When Design a counter that counts in the sequence: 5,4,3,2,1,0 and repeat , using JK flip flopsAfter you

COMPLETE the table below :

A	B	C	A+	B+	C+	JA	KA	JB	KB	JC	KC
0	0	0	1	0	1						
0	0	1	0	0	0						
0	1	0	0	0	1						
0	1	1									
1	0	0									
1	0	1									
1	1	0									
1	1	1									

Answer the following questions

1. The boolean function for JA is

Choose... ▾

2. The boolean function for KA is

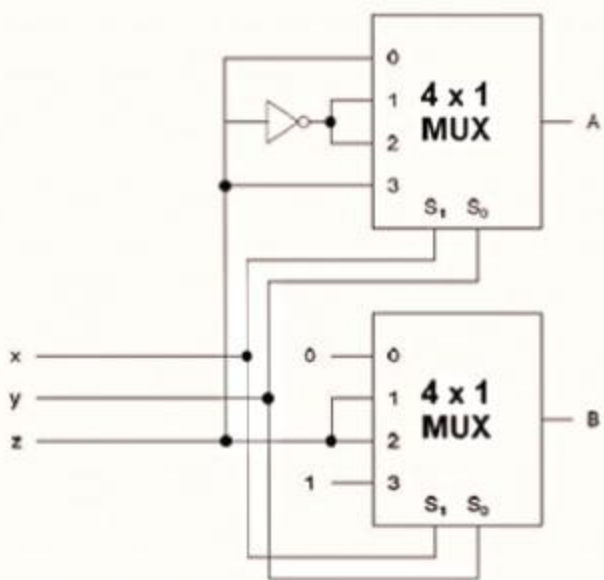
Choose... ▾

3. The boolean function for JC is

Choose... ▾

4. How many JK Flip Flop you must use to design the circuit?

▾



According to the circuit abovechoose the answer:

1. The output function " A " is expressed as

Choose...

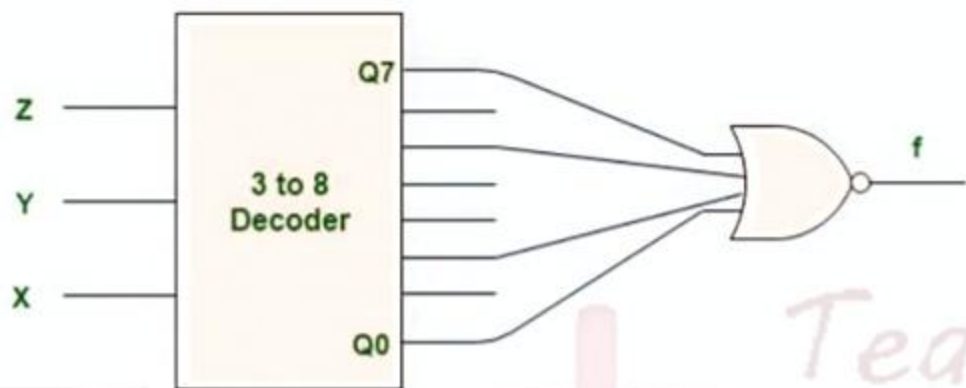
2. The output function " B " is expressed as

Choose...

3. The circuit shown above represents a

Choose...

What Boolean function does the circuit below realize ?



Select one:

- ☐ a. $x'y'+yz$
- ☐ b. $xz'+x'z$
- ☐ c. $xy+y'z'$
- ☐ d. $xz+x'z'$

Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

$f(x, y, a, b)$

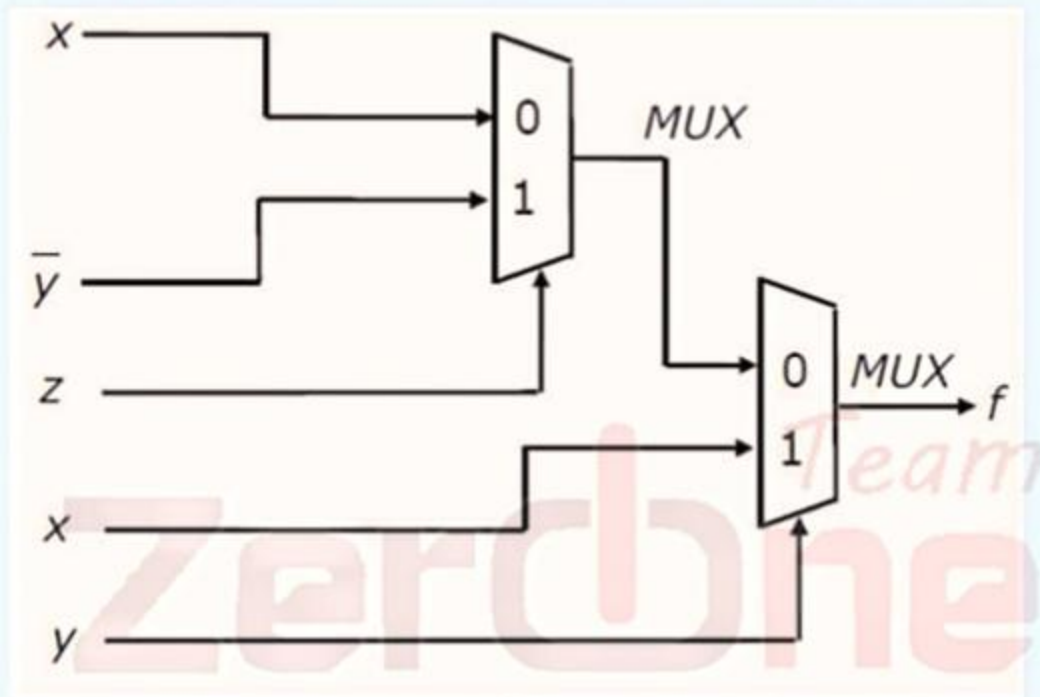
```
{  
  if (x is 1) y = a;  
  else y = b;  
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?

Select one:

- ☐ a. Decoder
- ☐ b. Flip Flop
- ☐ c. Multiplexer
- ☐ d. Full adder

Consider the circuit above. Which one of the following options correctly represents $f(x, y, z)$?

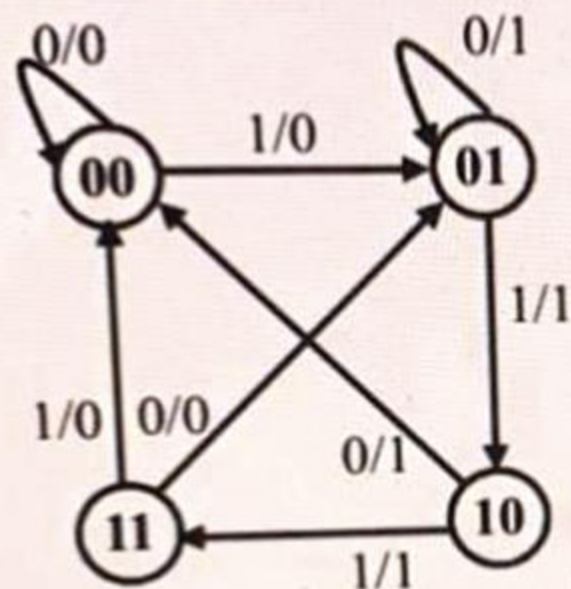


Select one:

- ☐ a. $xz' + xy + y'z$
- ☐ b. $xz' + xy + (yz)'$
- ☐ c. $xz + xy' + y'z$
- ☐ d. $xz + xy + (yz)'$

The state diagram below is for a sequential circuit that one input (X) and one output (Y) which of the following statement is TRUE :

hint # The sequence is X/Y

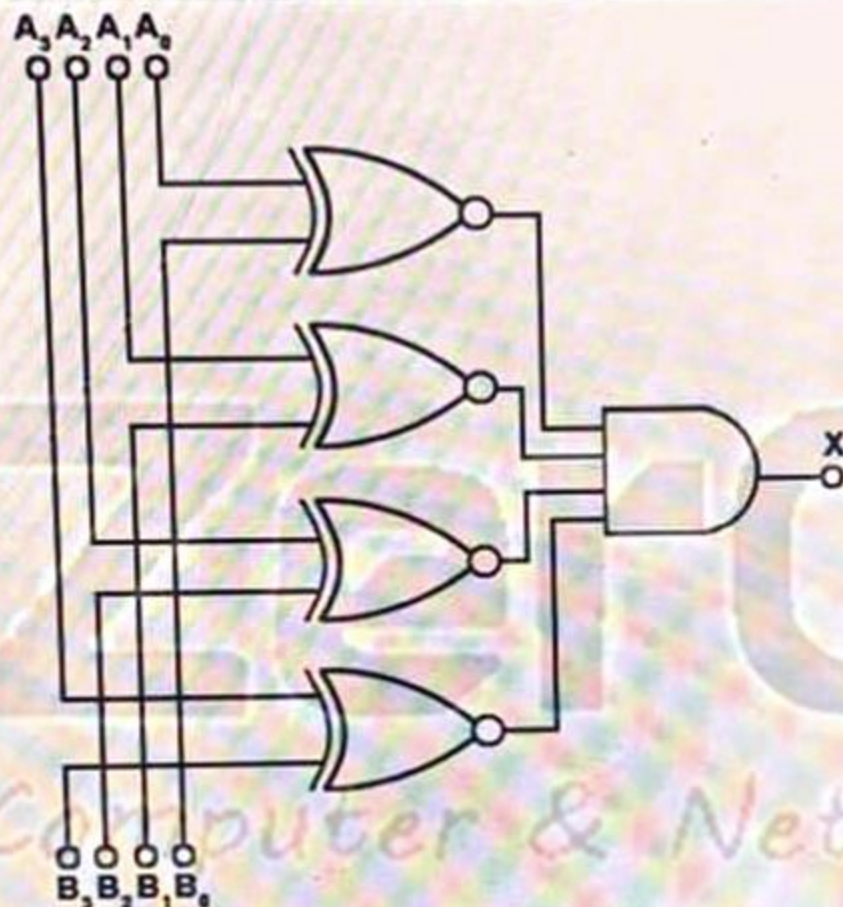


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Select one:

- ☐ a. Present state = Next state when x is 1 and $y = 0$
- ☐ b. Present state always = Next state when x is 0 and y is 1.
- ☐ c. None of all
- ☐ d. Next state does not depend on X NOR Y .

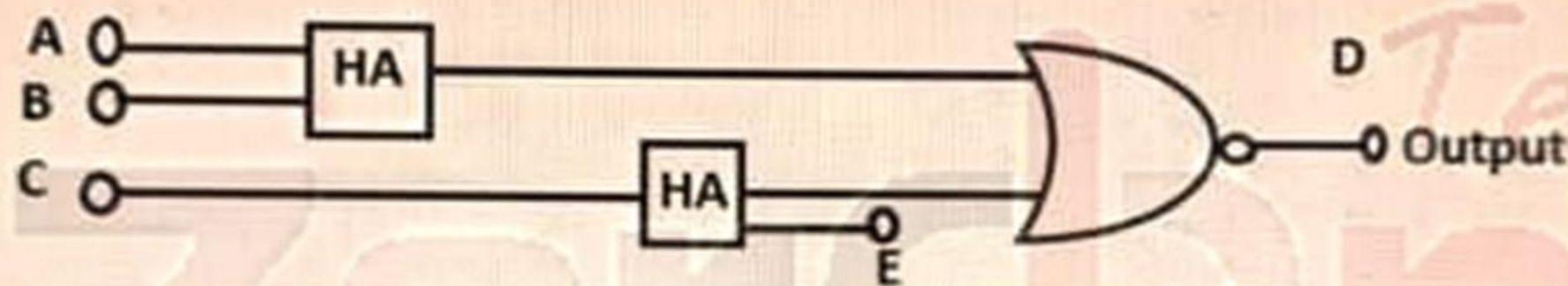
The figure below express a(n) state of 4 bit comparator



Select one:

- ☐ a. Less than
- ☐ b. Equality
- ☐ c. Greater Than

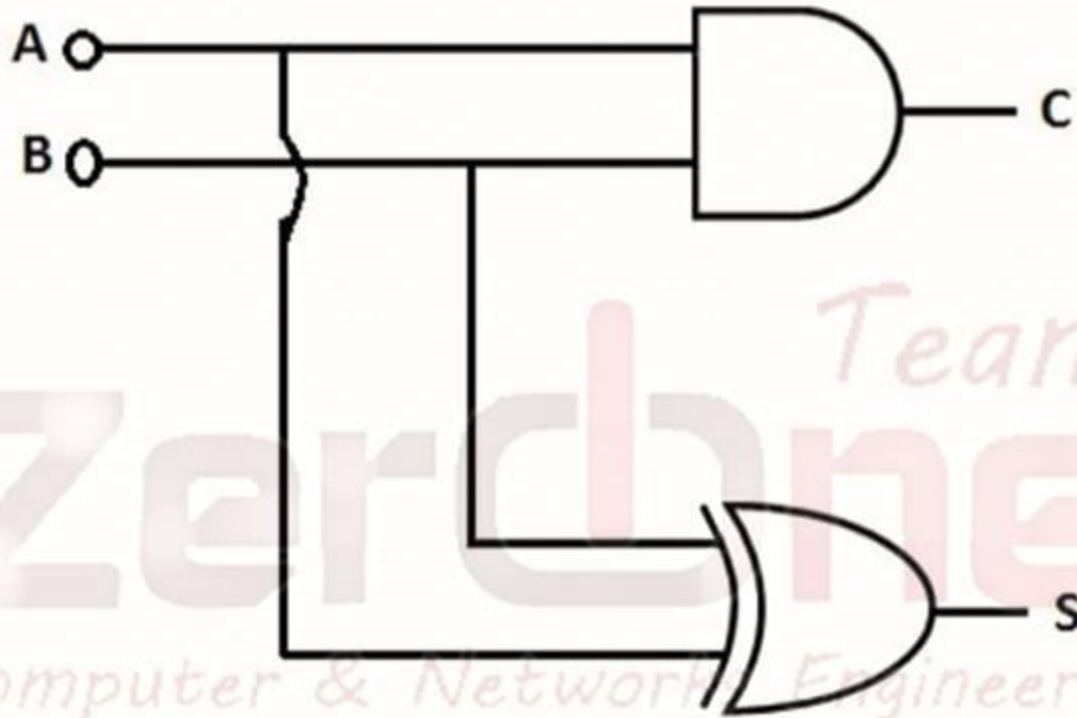
The circuit shown below is the



Select one:

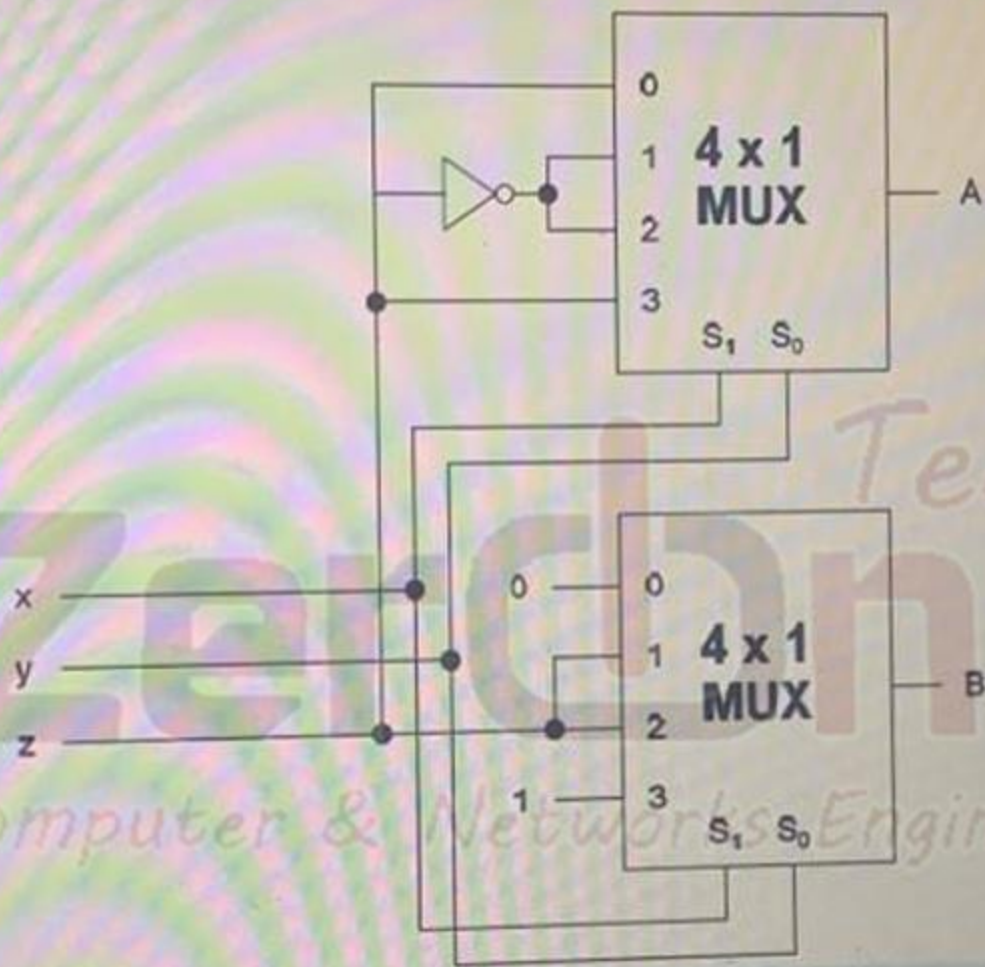
- ☐ a. Full subtractor
- ☐ b. Parity checker
- ☐ c. Full adder
- ☐ d. None of these

For the circuit shown for $AB = 00$, $AB = 01$, C, S values respectively are



Select one:

- ☐ a. 0, 0 and 1, 0
- ☐ b. 0, 0 and 0, 1
- ☐ c. 1, 0 and 0, 0
- ☐ d. 0, 1 and 0, 0



According to the circuit abovechoose the answer:

1. The output function "A" is expressed as

Choose... ▾

2. The output function "B" is expressed as

Choose... ▾

3. The circuit shown above represents a

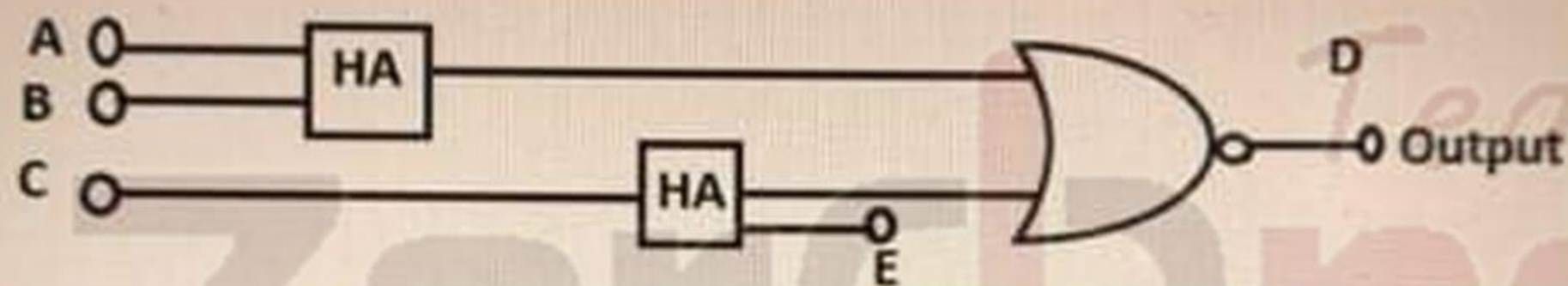
Choose... ▾

Which logic gates can be used to make a flip-flop?

Select one:

- ☒ a. NAND
- ☐ b. AND
- ☐ c. OR
- ☐ d. EXCLUSIVE OR

The circuit shown below is the



Select one:

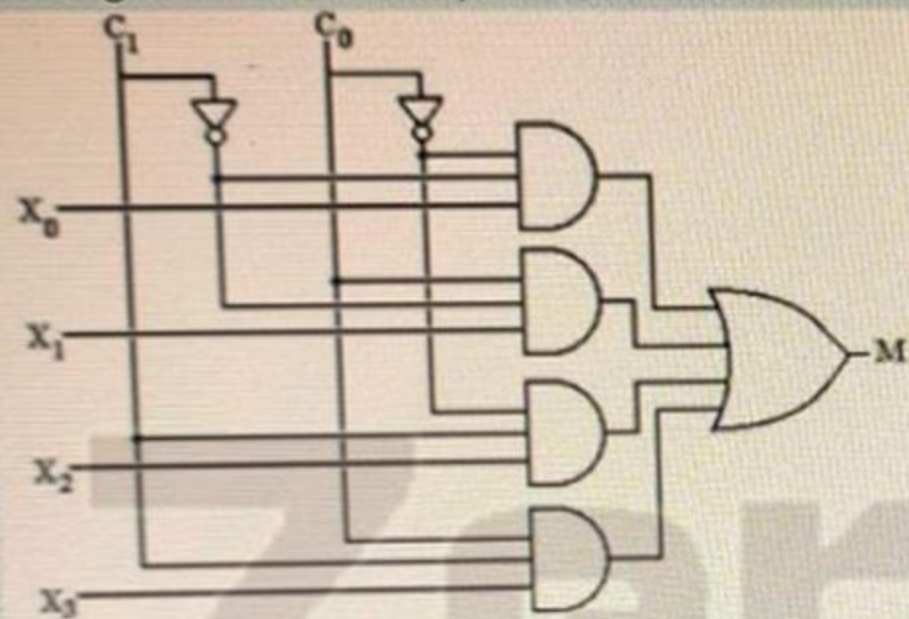
- ☐ a. Full subtractor
- ☐ b. Parity checker
- ☐ c. Full adder
- ☐ d. None of these

What is a sequential circuit?

Select one:

- ☐ a. a circuit composed of a sequence of elementary logic gates, e.g. AND gates
- ☐ b. a loop circuit, always repeating the same cycle
- ☐ c. a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element.
- ☐ d. a circuit in which the outputs depend only on the inputs at any moment

In the given 4-to-1 multiplexer, if $c_1 = 0$ and $c_0 = 1$ then the output M is _____



Select one:

- ☐ a. x_0
- ☐ b. x_3
- ☐ c. x_1
- ☐ d. x_2

How many (half adders) and (OR gates) are used to construct a full adder ?

Select one:

- ☐ a. 3 Half adders ,2 OR gate
- ☐ b. 2 Half adders ,1 OR gate
- ☐ c. 2 Half adders ,2 OR gate
- ☐ d. 2 Half adders ,3 OR gate

In a positive edge triggered SR flip-flop, a low S and low R produces

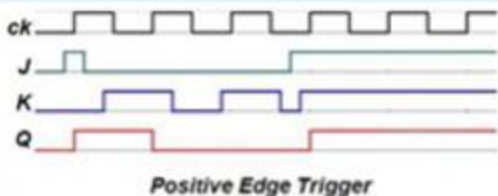
Select one:

- ☐ a. No change
- ☐ b. Toggle
- ☐ c. High state
- ☐ d. Low state

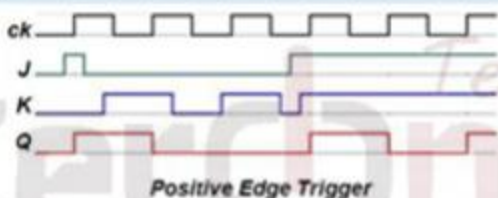
Which of the following Timing Diagram is Correct ,
if we use POSITIVE EDGE TRIGGER (J K) flip
flop :

Select one:

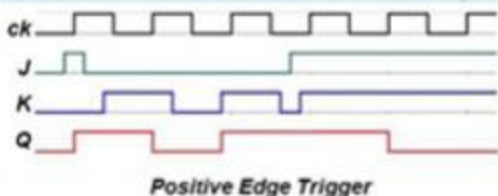
☐ a.



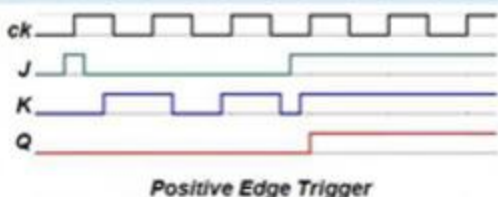
☐ b.



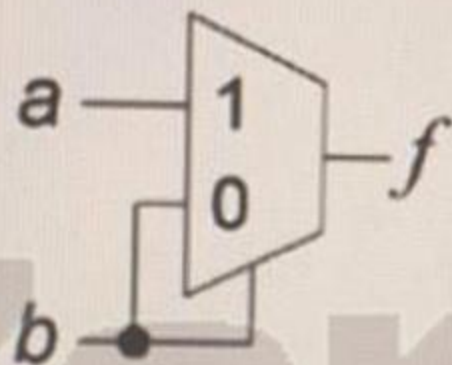
☐ c.



☐ d.



What logic function does this multiplexer circuit represent?



Select one:

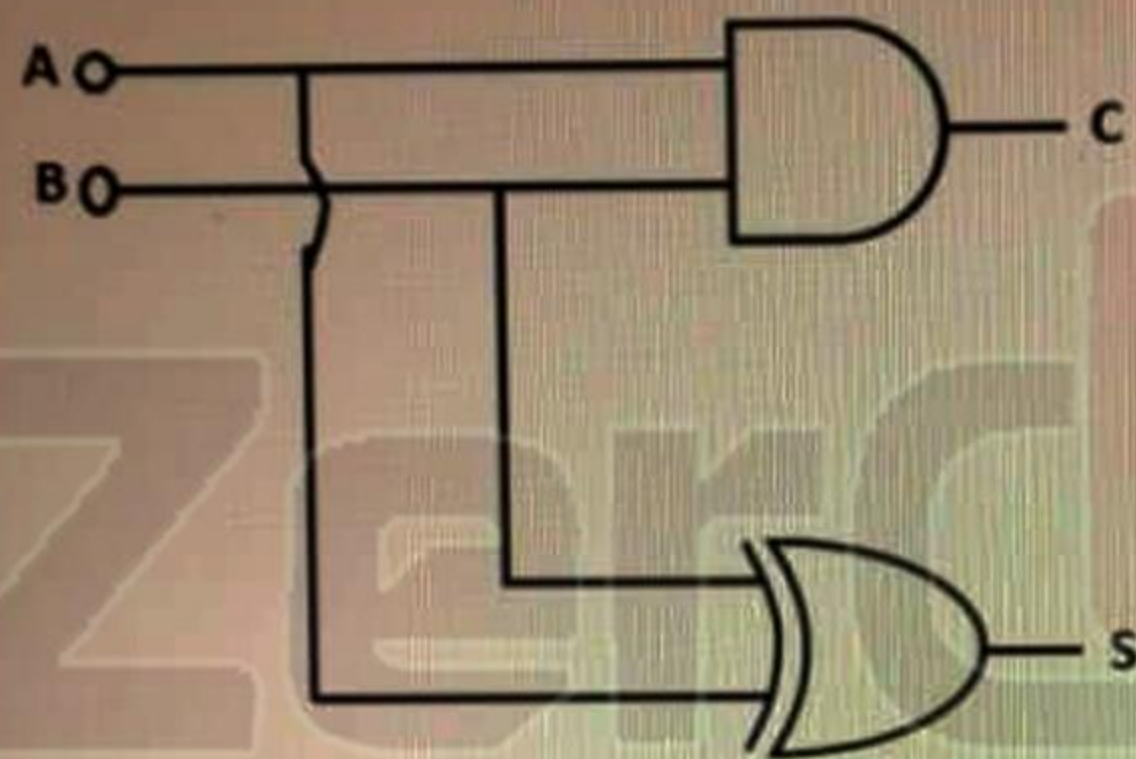
☒ a. $a' + b$

☐ b. $a.b'$

☐ c. $b'.a'$

☐ d. $b.a$

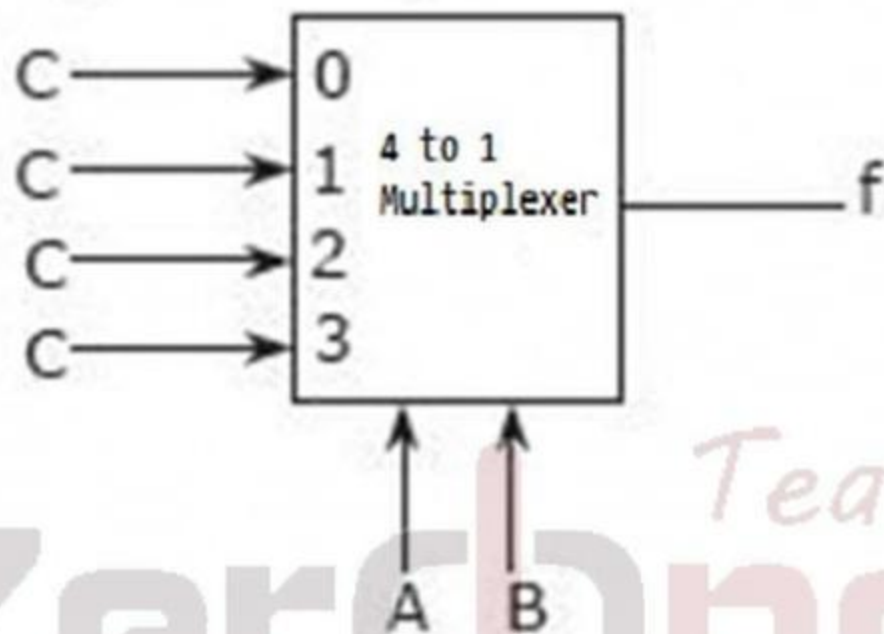
For the circuit shown for $AB = 00$, $AB = 01$, C, S values respectively are



Select one

- ☐ a. 1, 0 and 0, 0
- ☐ b. 0, 0 and 1, 0
- ☐ c. 0, 0 and 0, 1
- ☐ d. 0, 1 and 0, 0

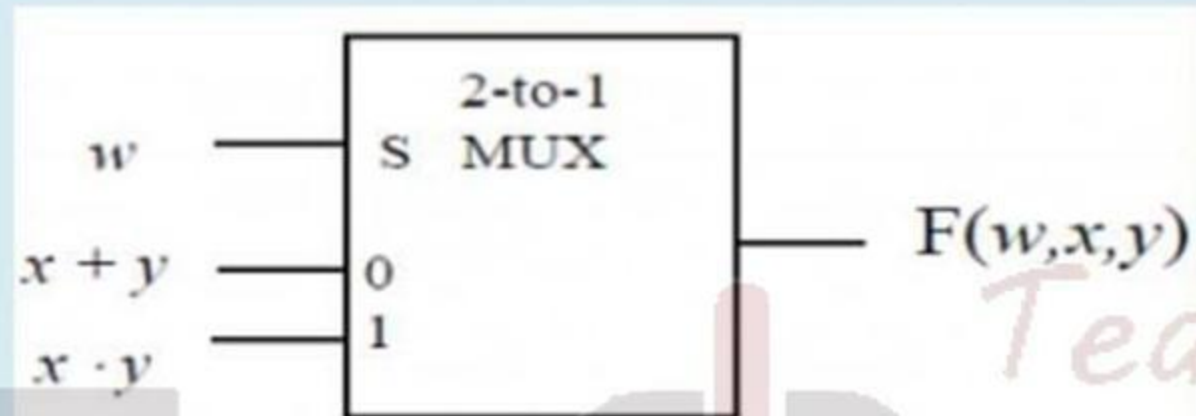
Consider the circuit in below figure , **F** implements
:



Select one:

- ☐ a. $A + B + C$
- ☐ b. $AB + BC + CA$
- ☐ c. $A \oplus B \oplus C$
- ☐ d. $(ABC)' + A'BC' + ABC$

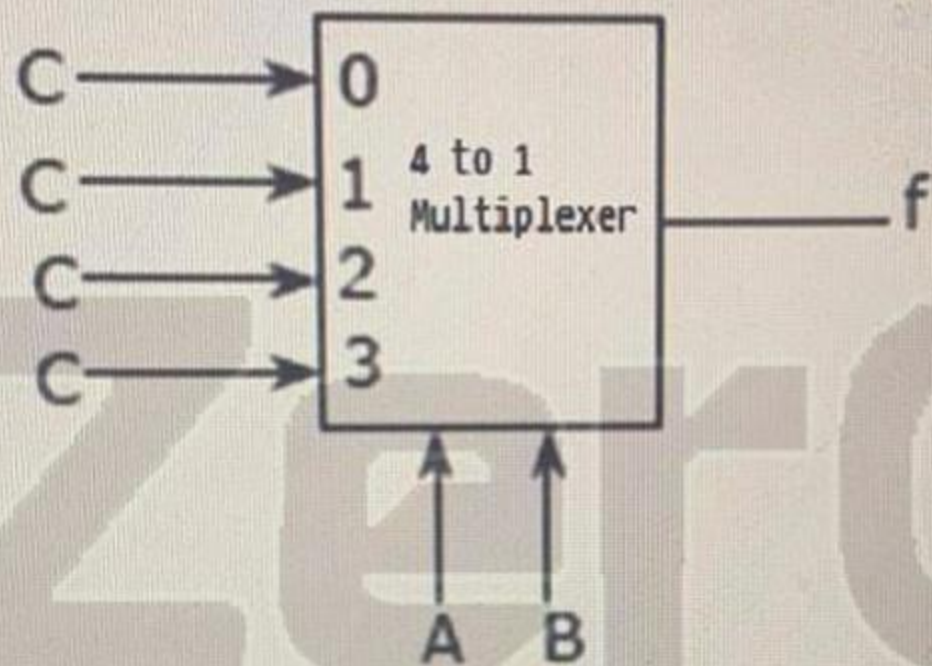
Find the boolean function $F(w,x,y)$:



Select one:

- ☐ a. $\sum m(2,3,7)$
- ☐ b. $\sum m(1,5,6,7)$
- ☐ c. $\sum m(1,2,5,6)$
- ☐ d. $\sum m(1,2,3,7)$

Consider the circuit in below figure , **F** implements :



Select one:

- ☐ a. $A + B + C$
- ☐ b. $A \oplus B \oplus C$
- ☐ c. $AB + BC + CA$
- ☐ d. $(ABC)' + A'BC' + ABC$

How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

Select one:

- ☐ a. 7
- ☐ b. 10
- ☐ c. 9
- ☒ d. 8

An 8 x 1 multiplexer has inputs A, B, and C connected to the selection inputs S_2 , S_1 , and S_0 , respectively. The data inputs, I_0 through I_7 , are as follows: $I_1 = I_2 = I_7 = 0$; $I_3 = I_5 = 1$; $I_0 = I_4 = D$; and $I_6 = D'$.

It implement the following function:

Select one:

- ☐ a. $F = \sum m(1, 6, 7, 9, 10, 11, 12, 13)$
- ☐ b. $F = \sum m(1, 6, 7, 9, 10, 11, 15)$
- ☒ c. $F = \sum m(1, 6, 7, 9, 10, 11, 12)$
- ☐ d. $F = \sum m(1, 6, 8, 9, 10, 11, 12)$

A combination circuit is specified by the following three Boolean functions.

$$F1(A, B, C) = \sum m(2, 4, 7)$$

$$F2(A, B, C) = \sum m(0, 3)$$

$$F3(A, B, C) = \sum m(0, 2, 3, 4, 7)$$

Select one:

- ☐ a. can be implemented by using decoder with NAND gates connected to the decoder outputs
- ☒ b. can be implemented by using decoder with AND gates connected to the decoder outputs
- ☐ c. can be implemented by using decoder with OR gates connected to the decoder outputs
- ☐ d. can be implemented by using decoder with NOR gates connected to the decoder outputs

When it is important to use a Tri-State buffer?

Select one:

- ☐ a. When all outputs are normally LOW
- ☐ b. When all outputs are normally HIGH
- ☐ c. When two or more outputs are connected to the same input
- ☐ d. When two or more outputs are connected to two or more inputs

How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

Select one:

- a. 7
- b. 10
- c. 9

A device that converts from decimal to binary numbered is called : -----

Select one:

- ☐ a. Decoder
- ☐ b. CPU
- ☐ c. Multiplexer
- ☒ d. Encoder

The following switching functions are to be implemented using a Decoder:

$$F_1 = \sum m(1, 2, 4, 8, 10, 14), F_2 = \sum m(2, 5, 9, 11), F_3 = \sum m(2, 4, 5, 6, 7)$$

The minimum configuration of the decoder should be

Select one:

- ☐ a. 2 – to – 4 lines.
- ☐ b. 3 – to – 8 lines
- ☐ c. 5 – to – 32 lines
- ☐ d. 4 – to – 16 lines.